ChessInsight: Visual Exploration of Gameplay Dynamics

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▲ MOTIVATION & PROBLEM

What is the problem?
Tools like Lichess and Chess.com analyse one game at a time via engine evaluation. They cannot reveal *population-level patterns* — how 350 K games worth of time usage, blunder rates, and position complexity differ across skill tiers.

Why does it matter?
Players, coaches, and researchers lack a visual system to understand *why* stronger players behave differently — e.g., how they allocate time across opening / middlegame / endgame or recover from time pressure.

Our goal
Build an **interactive visual analytics dashboard** that:
- Predicts skill tier from *behavioral* features (not raw Elo)
- Clusters players into interpretable behavioral archetypes
- Lets users explore patterns through linked, filterable views

▲ DATASET & FEATURES

Source: Lichess Open Database (data_1m_games.pgn)

Scale:
- **350,060** rapid / classical games
- **238,200** unique players in raw PGN
- **22,725** players with stable behavioral vectors (≥5 games)
- Elo range: **600 – 3,265** (beginner → titled)

Game-level features (47):
- Phase-wise avg. move time & time variance (opening / middlegame / endgame)
- Time-pressure frequency, low-time move ratio
- Centipawn loss, blunder & mistake rates
- Position complexity, opening aggression, piece activity

Player-level features (39):
- Aggregated means, variances, and derived ratios per player
- Time-accuracy interaction, error-severity score

▲ METHODS

Task 1 — Skill-Tier Classification
Predict 4 tiers (Beginner / Intermediate / Advanced / Expert) from behavioral features — not Elo.

- Models: Random Forest, XGBoost, soft-voting Ensemble
- SMOTE + class-weighted loss for imbalance
- Engineered features: phase-time ratios, time-pressure score, complexity–time interaction, error-severity
- Split: 230 K train / 49 K val / 49 K test

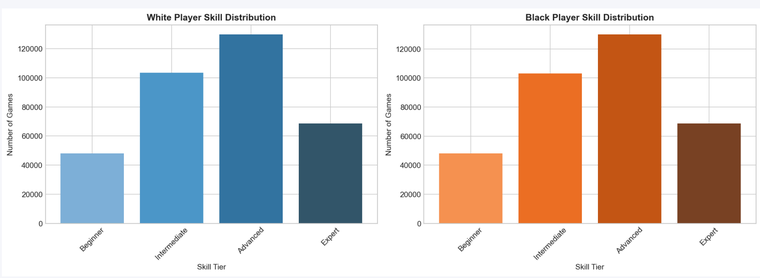
Task 2 — Behavioral Clustering
Discover recurring player archetypes without using Elo as a label.

- Algorithms: K-Means, BIRCH, Gaussian Mixture
- Sweep k = 3–7; select via Silhouette, Calinski–Harabasz, Davies–Bouldin
- Chosen: BIRCH, k = 5

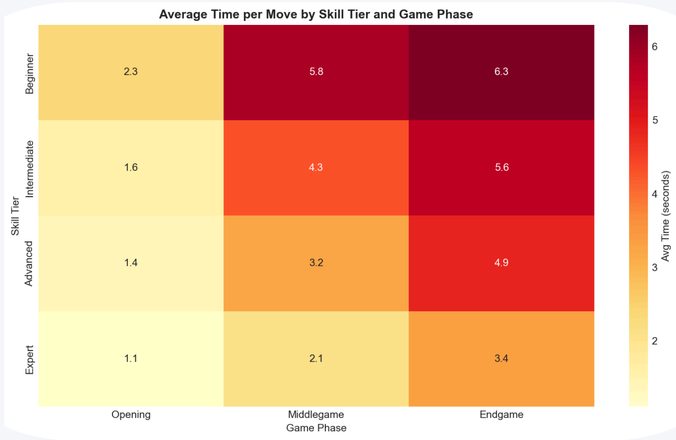
Dashboard (Streamlit)
- 2-D player map (PCA/t-SNE), phase-time heatmap, cluster archetype panels, confusion matrix — all linked by filters

▲ INTERACTIVE VISUALIZATIONS

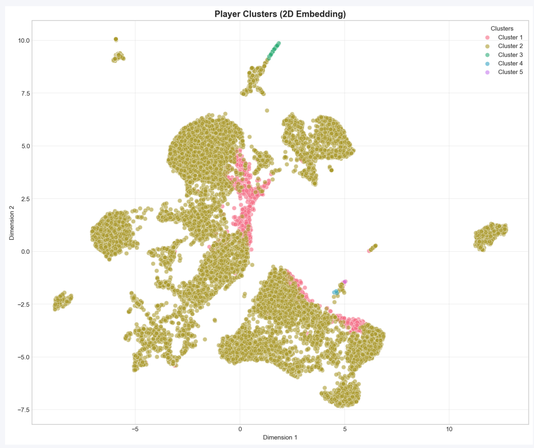
■ Skill Distribution
Bar chart of game counts across Beginner / Intermediate / Advanced / Expert tiers. Advanced dominates (124 K games), while Beginner is the smallest class (42 K). This class imbalance motivates SMOTE resampling in the classifier.



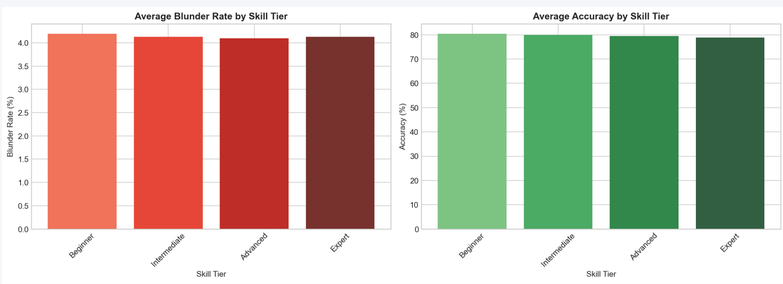
■ Phase-wise Time Heatmap
Heatmap of average seconds-per-move by skill tier × game phase. Stronger players invest proportionally more time in the middlegame; beginners often burn time early or move impulsively under pressure.

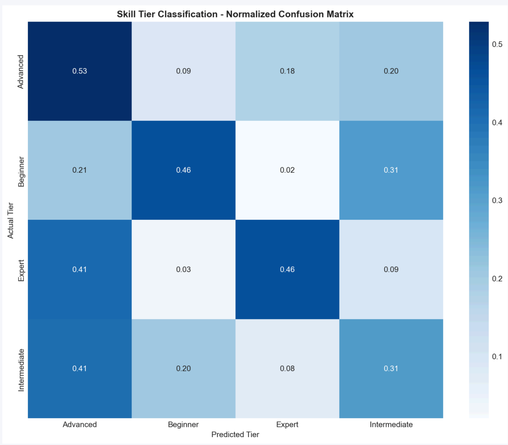


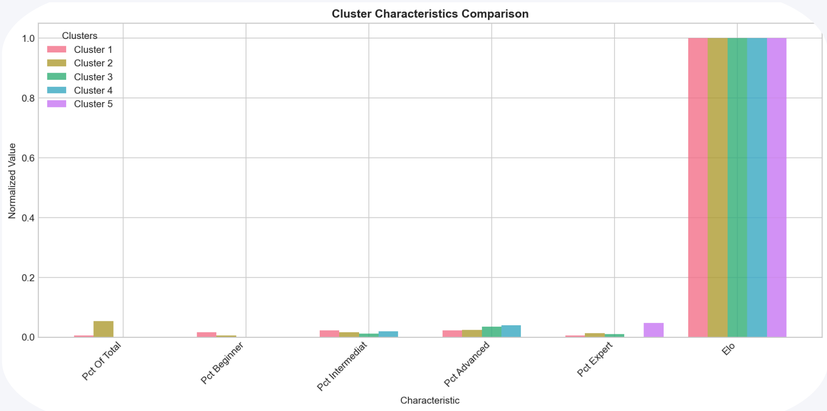
■ Player Behavior Map
2-D t-SNE / PCA scatter of 22 K players colored by cluster archetype. Reveals that behavioral similarity does not always align with Elo — some low-rated players cluster with higher-rated ones due to disciplined time management.



▲ RESULTS







44.5%	55.6%	0.44	0.296	5
Exact-Tier Test Accuracy	Adjacent-Tier Accuracy	Macro F1 Score	Silhouette Score (BIRCH k=5)	Behavioral Archetypes
XGBoost · 4-class · chance = 25%	Predictions within ±1 tier	Balanced across all tiers	Moderate cluster separation	Solid · Deliberate · Fast · Specialist · Elite

Skill Classification: XGBoost achieves 44.5% exact-tier and 55.6% adjacent-tier test accuracy — well above the 25% random baseline for a 4-class task. Macro F1 ≈ 0.44 confirms balanced performance across tiers. The most important features are *middlegame avg. move time* (0.16), *opening time variance* (0.12), and *position complexity* (0.10). Most misclassifications occur between adjacent tiers (Intermediate ↔ Advanced, Advanced ↔ Expert), showing the model has learned a sensible skill ordering.

Behavioral Clustering: BIRCH with k = 5 yields silhouette = 0.296, CH-index = 1826, DB-index = 1.04 — the best trade-off across methods tested. Five distinct archetypes emerge: **Solid Advanced Core** (21 K players, avg. Elo 1714), **Deliberate Intermediates** (1.6 K, avg. Elo 1505), **High-Level Deliberate Specialists** (74 players, avg. Elo 1780), **Fast Advanced Grinders** (3 players), and an **Elite Expert** outlier (avg. Elo 2119).

▲ CONCLUSIONS & FUTURE WORK

Conclusions
- ChessInsight scales behavioral analysis to **350 K games** and **22 K players**, far beyond what any single-game engine can show.
- Time management — specifically phase-wise move-time allocation — is the **strongest behavioral predictor of skill**, outweighing blunder rates and piece activity.
- Exact-tier accuracy of 44.5% and adjacent accuracy of 55.6% confirm that behavioral features carry significant, but partial, skill signal. There is genuine ambiguity at tier boundaries.
- BIRCH clustering reveals **5 interpretable archetypes**, demonstrating that multiple behavioral pathways lead to similar Elo — coaching is not one-size-fits-all.
- The Streamlit dashboard connects all components into a single interactive experience for players, coaches, and researchers.

Limitations
- Tier boundaries are soft: Intermediate ↔ Advanced confusion is high.
- Cluster sizes are highly unequal; smallest clusters have < 5 players and cannot be reliably interpreted.
- Analysis limited to games with reliable clock data — blitz omitted.

Future Work
- Add **move-by-move time series** features and opening-family behavioral signatures.
- Explore semi-supervised and density-based clustering to stabilize archetypes.
- Calibrate predicted skill tier against future rating gain as a longitudinal ground truth.
- Deploy dashboard publicly on Lichess game histories and conduct user studies to measure coaching impact.

Data: Lichess Open Database · Tools: Python, Streamlit, Plotly, scikit-learn, XGBoost, Stockfish ·